A Taxonomy of Hallucination Types in Large Language Models for Research Applications

Abstract

Large language models (LLMs) are increasingly embedded in scholarly workflows, from literature synthesis to citation generation and exploratory data analysis, yet they remain prone to producing fluent but unsupported or false content commonly termed "hallucination." The heterogeneity of this phenomenon has produced a proliferating and sometimes inconsistent vocabulary across the literature, complicating efforts to detect, mitigate, and communicate hallucination risk in research settings. This paper synthesizes existing taxonomic work (Huang et al., 2025; Ji et al., 2023; Zhang et al., 2023) into a taxonomy oriented toward research applications, distinguishing hallucinations along four axes: origin (intrinsic versus extrinsic), evaluative target (factuality versus faithfulness), content type (entity, citation, numerical, methodological, and logical hallucinations), and task context (closed-book generation, retrieval-augmented generation, and long-form multi-step synthesis). We review empirical evidence on the prevalence of citation and reference fabrication in academic use (Walters & Wilder, 2023; Dahl et al., 2024), survey current detection and mitigation approaches including retrieval augmentation, semantic-entropy-based uncertainty estimation, and sampling-based consistency checking, and identify persisting challenges around benchmark validity, cross-domain transfer, and the absence of a standardized taxonomy for scholarly use cases. We conclude by outlining research gaps and directions for building hallucination-aware tools suited to research and academic writing contexts.

Keywords: large language models; hallucination taxonomy; factuality; faithfulness; citation fabrication; retrieval-augmented generation; research integrity; natural language generation

1. Introduction

The rapid diffusion of large language models (LLMs) such as GPT-4, Claude, Gemini, and their successors into scholarly practice has created new opportunities for accelerating literature review, drafting, data summarization, and code generation. It has also surfaced a persistent and consequential failure mode: LLMs frequently generate content that is fluent, internally coherent, and superficially authoritative, yet factually incorrect or unsupported by any real source. This behavior is widely termed "hallucination," a metaphor borrowed from perceptual psychology to describe outputs that resemble knowledge without being grounded in it (Ji et al., 2023).

Hallucination is not a single, uniform error type. It ranges from small numerical slips to the wholesale invention of nonexistent bibliographic references, legal precedents, or experimental results. Multiple research groups have proposed taxonomies to organize this diversity (Huang et al., 2025; Rawte et al., 2023; Ye et al., 2023; Zhang et al., 2023), but these taxonomies were developed primarily for general natural language generation (NLG) or open-domain question answering, and their terminology is not always consistent across papers. For research applications specifically—literature synthesis, citation generation, data interpretation, and scholarly writing assistance—the practical consequences of different hallucination types differ substantially, and a taxonomy that maps clearly onto these consequences is needed to guide detection tool design, user training, and editorial policy.

This paper has three aims. First, it synthesizes the major existing taxonomies of LLM hallucination into a coherent framework, reconciling overlapping terminology. Second, it extends this framework with a content-type dimension specifically relevant to research use, distinguishing entity, citation, numerical, methodological, and logical hallucinations, and grounds this dimension in empirical studies of citation fabrication (Dahl et al., 2024; Walters & Wilder, 2023). Third, it reviews current detection and mitigation approaches, identifies open challenges, and proposes directions for future work targeted at research and academic-writing applications of LLMs.

The remainder of the paper is organized as follows. Section 2 reviews related work on hallucination definitions and taxonomies. Section 3 presents the proposed taxonomy across four axes. Section 4 surveys current detection and mitigation approaches. Section 5 discusses research challenges and limitations. Section 6 identifies research gaps and future directions, and Section 7 concludes.

2. Background and Related Work

Systematic study of hallucination predates the LLM era. In abstractive summarization, Maynez et al. (2020) introduced an influential distinction between intrinsic hallucinations, in which generated content directly contradicts the source document, and extrinsic hallucinations, in which generated content cannot be verified from the source at all, whether or not it is true. This intrinsic/extrinsic distinction remains foundational to later LLM taxonomies, even as the notion of a bounded “source document” has become more ambiguous for open-domain generation.

Ji et al. (2023), in a widely cited survey published in ACM Computing Surveys, generalized hallucination research across NLG tasks and proposed a two-way distinction between factuality hallucination (inconsistency with verifiable real-world facts) and faithfulness hallucination (inconsistency with source content, user instructions, or the model's own internal logic). This factuality/faithfulness distinction has become a standard organizing principle for later work.

With the emergence of general-purpose LLMs, several surveys extended and refined this picture. Huang et al. (2025), in a survey originally circulated as a preprint and later published in ACM Transactions on Information Systems, propose an “innovative taxonomy” that subdivides factuality hallucination into factual inconsistency and factual fabrication, and faithfulness hallucination into instruction inconsistency, context inconsistency, and logical inconsistency; they further examine contributing factors across the data, training, and inference stages of the LLM pipeline. Zhang et al. (2023), in the “Siren’s Song” survey, propose an overlapping but differently labeled taxonomy distinguishing input-conflicting, context-conflicting, and fact-conflicting hallucinations, and connect these categories to mechanisms rooted in pretraining data quality, alignment tuning, and decoding strategies. Rawte et al. (2023) instead propose an orthogonal axis contrasting “factual mirage” and “silver lining” hallucinations, again layered onto an intrinsic/extrinsic split. Ye et al. (2023) focus specifically on hallucination that arises during multi-step reasoning, terming their review “Cognitive Mirage” and cataloguing how errors compound across reasoning chains. Tonmoy et al. (2024) complement these taxonomic surveys with a comprehensive review of mitigation techniques spanning prompting, retrieval augmentation, fine-tuning, and self-refinement.

A separate empirical literature has quantified hallucination in domains directly relevant to research and professional practice. Dahl et al. (2024) developed a typology of legal hallucinations and found that general-purpose LLMs produced legally inconsistent or fabricated case information between 58% (GPT-4) and 88% (Llama 2) of the time when queried about verifiable federal court cases. In the domain of scholarly citation, Walters and Wilder (2023) found that ChatGPT-3.5 fabricated 55% of bibliographic citations in generated literature reviews, a rate that fell to 18% for ChatGPT-4, while a substantial share of even the non-fabricated citations contained substantive errors—demonstrating that citation hallucination, though model-dependent, remains a persistent risk even in more capable systems.

Alongside taxonomic and empirical work, a parallel literature has developed benchmarks and detection methods. TruthfulQA (Lin et al., 2022) measures whether models reproduce common human misconceptions; HaluEval (Li et al., 2023) provides a large-scale benchmark of human-annotated hallucinated and non-hallucinated responses; and FActScore (Min et al., 2023) proposes atomic-fact-level evaluation of long-form factual precision. Detection methods such as SelfCheckGPT (Manakul et al., 2023) and semantic-entropy-based uncertainty estimation (Farquhar et al., 2024) attempt to flag likely hallucinations without requiring external ground truth, discussed further in Section 4. Finally, Kalai et al. (2025) offer a theoretical account situating hallucination not as an idiosyncratic bug but as a natural consequence of training and evaluation procedures that reward confident guessing over calibrated uncertainty, a perspective that reframes hallucination as a statistically predictable outcome of current LLM development pipelines rather than a defect to be fully eliminated by better prompting alone.

Taken together, this body of work establishes that (a) hallucination is multidimensional, (b) its terminology has not fully converged across surveys, and (c) empirical prevalence varies sharply by domain and task. The taxonomy proposed in Section 3 draws on these strands but organizes them specifically around the concerns of research and scholarly-writing applications, where citation integrity, methodological accuracy, and traceable evidence are paramount.

3. A Taxonomy of Hallucination Types for Research Applications

We organize LLM hallucination along four complementary axes: (3.1) origin, (3.2) evaluative target, (3.3) content type, and (3.4) task context. These axes are not mutually exclusive; a single hallucinated output (for example, a fabricated citation supporting a false claim) can typically be classified along all four simultaneously. The value of separating them is diagnostic: origin and evaluative target help explain why a hallucination occurred, while content type and task context help predict where it is likely to occur and how costly it is likely to be in a research setting.

3.1 Origin-Based Distinction: Intrinsic versus Extrinsic Hallucination

Following Maynez et al. (2020), intrinsic hallucinations occur when generated text actively contradicts an available source—for example, an LLM asked to summarize a paper's methodology but describing a different statistical test than the one the paper actually used. Extrinsic hallucinations occur when generated text introduces content that cannot be verified against any available source, regardless of its truth value—for instance, adding a plausible-sounding but untraceable claim about a study's sample size. In research applications, intrinsic hallucination is especially dangerous during literature summarization and evidence synthesis, since it misrepresents a real, checkable source, while extrinsic hallucination is the dominant mode in citation and reference generation, where the model invents content with no source at all (Dahl et al., 2024; Walters & Wilder, 2023).

3.2 Evaluative Target: Factuality versus Faithfulness Hallucination

Building on Ji et al. (2023) and the refinement offered by Huang et al. (2025), factuality hallucination refers to inconsistency between generated content and verifiable real-world knowledge, and is further divided into factual inconsistency (contradicting known facts) and factual fabrication (inventing facts with no real-world referent). Faithfulness hallucination refers to inconsistency between generated content and the user's instructions, supplied context, or the model's own preceding reasoning, subdivided into instruction inconsistency, context inconsistency, and logical inconsistency. For research applications, factuality hallucination governs whether a claim about the world is true, while faithfulness hallucination governs whether the model accurately represents the specific documents, data, or instructions a researcher has provided—a distinction with direct implications for tool design, since retrieval augmentation primarily targets factuality gaps while context-grounding and prompt engineering primarily target faithfulness gaps.

3.3 Content-Type Taxonomy for Research Use

Because generic NLG taxonomies do not differentiate content types with the granularity that scholarly use requires, we propose the following content-type categories, informed by empirical studies of citation and legal hallucination:

- Entity and attribute hallucination: invented or misattributed names, dates, institutional affiliations, or credentials, including author misattribution.
- Citation and reference hallucination: fabricated bibliographic entries, including nonexistent titles, journals, or DOIs, or citations that superficially resemble real entries but link to unrelated work. Walters and Wilder (2023) found fabrication rates of 55% (GPT-3.5) and 18% (GPT-4) across 636 generated citations, with many fabricated entries including plausible but invalid or misdirected DOIs.
- Numerical and statistical hallucination: invented sample sizes, effect sizes, p-values, or other quantitative results not present in any source.
- Methodological and procedural hallucination: fabricated descriptions of experimental design, analytic pipelines, or procedures that were never actually performed, sometimes presented with the structure of an empirical study.
- Logical and reasoning hallucination: internally inconsistent or invalid inference chains that nonetheless appear coherent on the surface, particularly prevalent in multi-step reasoning tasks (Ye et al., 2023).

- Source-attribution hallucination: correctly stating a claim that does exist in the literature but attributing it to the wrong source, author, or year, a hybrid failure that is especially difficult for readers to detect because the underlying claim is factually true.

Domain-specific studies illustrate how these categories manifest at scale. In the legal domain, Dahl et al. (2024) documented case law hallucinations combining fabricated citations, invented holdings, and misattributed court levels, with hallucination rates varying systematically by court hierarchy and jurisdiction. In scholarly writing, citation hallucination has been shown to interact with topic familiarity: fabrication is markedly higher for less-studied or narrower subfields than for extensively documented topics, suggesting that content-type hallucination risk is not uniform but tracks the density of relevant material in a model's training distribution.

3.4 Task-Context Taxonomy

Hallucination risk also varies systematically with the generation context in which a model operates. Closed-book or purely parametric generation, in which the model answers from its internal parameters without external grounding, carries the highest risk of factuality and citation hallucination, consistent with the account offered by Kalai et al. (2025), who argue that hallucination arises predictably when training and evaluation incentives reward confident guessing over calibrated abstention. Retrieval-augmented generation (RAG), in which the model is supplied with retrieved documents at inference time (Lewis et al., 2020), reduces but does not eliminate hallucination: models may still misrepresent retrieved content (an intrinsic, faithfulness-type failure) or introduce claims beyond what was retrieved (an extrinsic, factuality-type failure). Long-form and multi-step synthesis tasks, such as generating an extended literature review or a multi-stage analysis, introduce a further compounding risk: errors introduced at one step can propagate and be treated as established fact in subsequent steps, a pattern documented in reasoning-chain hallucination research (Ye et al., 2023) and consistent with the higher fabrication rates observed in longer, more structurally complex outputs such as full literature reviews compared to short factual answers.

4. Current Approaches and Developments

4.1 Detection

Detection approaches fall broadly into benchmark-based evaluation, sampling-based consistency checking, and uncertainty-based estimation. Benchmark-based approaches evaluate model outputs against curated ground truth: TruthfulQA (Lin et al., 2022) tests whether models reproduce common human misconceptions across 38 categories; HaluEval (Li et al., 2023) provides a large-scale benchmark of human- and model-annotated hallucinated content across question answering, summarization, and dialogue; and FActScore (Min et al., 2023) decomposes long-form generations into atomic facts and scores each against a knowledge source, offering finer-grained precision estimates than sentence- or document-level judgments.

Sampling-based approaches instead exploit the intuition that hallucinated content tends to be less consistent across repeated generations than well-grounded content. SelfCheckGPT (Manakul et al., 2023) operationalizes this by sampling multiple responses to the same query and measuring their mutual consistency without requiring access to model internals or an external knowledge base, making it applicable as a black-box detector. Uncertainty-based approaches move a step further by formalizing this intuition statistically: Farquhar et al. (2024), in a paper published in Nature, introduce semantic entropy, which clusters sampled generations by meaning-level equivalence rather than surface wording and computes entropy over these semantic clusters, enabling detection of a specific hallucination subtype the authors term “confabulation”—arbitrary, incorrect generation—without requiring labeled ground truth or task-specific supervision.

4.2 Mitigation

Mitigation strategies broadly separate into architectural, decoding-time, and training-time interventions, comprehensively reviewed by Tonmoy et al. (2024). Architecturally, retrieval-augmented generation (Lewis et al., 2020) grounds generation in retrieved non-parametric memory, generating more specific and verifiable output than parametric-only generation, and remains the dominant strategy for research tools that must cite traceable sources. Decoding-time strategies include prompting techniques that encourage step-by-step verification or explicit citation of retrieved passages, and self-consistency or self-refinement loops in which a model critiques and revises its own output. Training-time strategies include fine-tuning on hallucination-annotated data and reinforcement-learning-based alignment toward calibrated uncertainty. Kalai et al. (2025) argue that a socio-technical shift in evaluation design—explicitly rewarding models for expressing appropriate uncertainty or abstaining rather than guessing—is necessary because current benchmark scoring conventions structurally incentivize confident hallucination over calibrated honesty.

4.3 Domain-Specific Tooling for Research

Beyond general-purpose detection and mitigation, research-specific tooling has begun to target citation and reference integrity directly, including DOI cross-verification, citation-existence checking against bibliographic databases, and retrieval-grounded citation generation that constrains outputs to sources actually retrieved. However, evaluations in specialized professional domains show that even purpose-built, retrieval-augmented legal research tools have not eliminated hallucination, underscoring that grounding alone is a necessary but not sufficient condition for reliability (Dahl et al., 2024).

5. Research Challenges and Limitations

Several challenges limit both the study and mitigation of hallucination in research applications. First, taxonomic inconsistency across the literature—with overlapping but differently labeled categories from Huang et al. (2025), Zhang et al. (2023), Rawte et al. (2023), and Ye et al. (2023)—complicates cross-study comparison and slows convergence toward standardized evaluation protocols. Second, existing benchmarks such as TruthfulQA were designed to probe adversarially chosen misconceptions and may not represent the distribution of errors encountered in realistic research workflows such as literature synthesis or citation generation, raising construct validity concerns when such benchmarks are used to certify tools for scholarly use.

Third, current detection methods carry practical costs and coverage limitations. Sampling-based and semantic-entropy-based methods (Manakul et al., 2023; Farquhar et al., 2024) require multiple generations per query, increasing inference cost, and are best validated on short-form factual question answering rather than the long-form, multi-claim outputs typical of literature reviews or research summaries. Fourth, hallucination prevalence and type vary substantially by domain and topic familiarity: citation fabrication rates documented by Walters and Wilder (2023) and legal hallucination rates documented by Dahl et al. (2024) differ by tens of percentage points depending on subject matter, jurisdiction, and model version, meaning that findings from one domain cannot be assumed to generalize to another. Fifth, human verification remains necessary even with current mitigation strategies; retrieval augmentation and self-consistency checks reduce but do not eliminate hallucination, and no current method offers a reliability guarantee suitable for unsupervised scholarly use. Finally, most existing evaluation focuses on single-turn or single-document outputs, leaving compounding, multi-step hallucination in longer research-support workflows comparatively understudied (Ye et al., 2023).

6. Research Gaps and Future Directions

Building on the challenges above, we identify five directions for future work. First, the field would benefit from a standardized, cross-domain taxonomy specifically oriented to research and scholarly-writing use cases, reconciling the overlapping vocabularies of existing surveys (Huang et al., 2025; Ji et al., 2023; Zhang et al., 2023) with the content-type distinctions proposed in Section 3.3. Second, there is a need for benchmarks that directly measure citation fidelity and reference grounding in literature-review and citation-generation tasks, rather than relying on general factuality benchmarks such as TruthfulQA (Lin et al., 2022) that were not designed for this purpose.

Third, tighter integration of provenance-tracking and verifiable-generation techniques into retrieval-augmented pipelines used for research tools could allow every generated claim to be traced to a specific retrieved passage, making faithfulness violations mechanically detectable rather than requiring post hoc checking. Fourth, following the calibration-oriented account of Kalai et al. (2025), evaluation frameworks for research-oriented LLM tools should explicitly reward appropriate abstention and uncertainty expression, rather than implicitly penalizing models for declining to answer when evidence is insufficient. Fifth, longitudinal and discipline-specific evaluation is needed: because hallucination prevalence appears to vary with topic familiarity and domain structure (Dahl et al., 2024; Walters & Wilder, 2023), taxonomy validation and mitigation testing should be replicated across the humanities, social sciences, and natural sciences rather than generalized from any single domain. Finally, multimodal hallucination—misrepresentation of tables, figures, and numerical data extracted from source documents—remains comparatively underexplored relative to text-only hallucination and warrants dedicated taxonomic and empirical attention as LLM-based research tools increasingly ingest non-textual scholarly content.

7. Conclusion

Hallucination in large language models is not a single phenomenon but a family of related failures that differ in origin, evaluative target, content type, and task context. This paper synthesized existing taxonomic work (Huang et al., 2025; Ji et al., 2023; Maynez et al., 2020; Zhang et al., 2023) and extended it with a content-type dimension grounded in empirical studies of citation and legal hallucination (Dahl et al., 2024; Walters & Wilder, 2023), producing a taxonomy oriented specifically toward research and scholarly-writing applications. Current detection methods, including benchmark-based evaluation, sampling-based consistency checking, and semantic-entropy-based uncertainty estimation (Farquhar et al., 2024; Manakul et al., 2023; Min et al., 2023), together with mitigation strategies such as retrieval augmentation (Lewis et al., 2020), have measurably reduced but not eliminated hallucination risk. Given documented fabrication rates as high as 55% for bibliographic citations and 88% for legal case information in some model and domain combinations, the practical stakes for research integrity are substantial. Continued progress will depend on standardizing taxonomy across the field, developing benchmarks tailored to scholarly tasks, integrating provenance tracking into research-oriented generation pipelines, and reorienting evaluation incentives toward calibrated uncertainty rather than confident guessing. Until such measures mature, human verification of LLM-generated citations, facts, and methodological claims remains an indispensable safeguard for research integrity.

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